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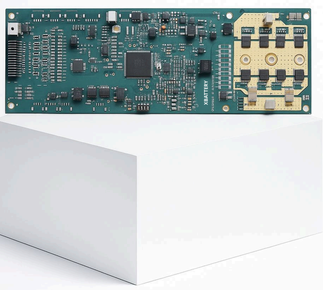
XBattery BharatBMS

An intelligent, scalable, high-voltage Battery Management System (BMS) supporting up to 800 volts for EVs and energy storage.



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XBattery BharatBMS-ESS-48V

The BharatBMS-ESS-48V redefines energy management with intelligent performance, robust safety, and scalable design.
Engineered for next-generation ESS solutions, it supports up to 18S LFP and NMC battery packs, ensuring precise monitoring, intelligent cell balancing, and multi-level protection, all in a compact, space-efficient form.
With seamless CAN, UART, and BLE connectivity, it delivers real-time insights, smart control, and optimized efficiency—offering reliability and long-term durability for modern energy systems.

Applications

Ideal for a wide range of energy solutions:

Renewable Energy Storage
Solar and wind power systems

Telecom Infrastructure
Reliable backup for towers

Electric Vehicles
Safe and efficient battery control

Backup & UPS Systems
Uninterrupted, stable energy support

Industrial Battery Packs
High-performance power systems

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Core Features

Intelligent Cell Monitoring
Supports up to 18 series cells with ±2 mV precision and passive balancing.

Communication & Control
CAN FD, UART, SPI, and Ethernet for real-time data and remote monitoring.

Safety First
ISO 26262-compliant protection with fault diagnostics for voltage, current, and temperature.

Thermal Management
Real-time temperature sensing and thermal runaway detection for efficient heat control.

Scalable Architecture
Modular design allows flexible series and parallel configurations.

Smart Diagnostics
Real-time visualization and lifecycle analytics for optimal performance.

Additional Features

SOC/SOH Algorithms
Accurate State-of-Charge and State-of-Health estimation.

Optimized Software
High-efficiency C framework for real-time applications.

Customizable API
Open API for seamless integration and control.

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Tech Specifications

General & Electrical Specifications	Parameters
Supported Cell Chemistry	LFP/NMC
Cell Count Range	13S to 17S
Nominal Pack Voltage / Range	42.9V to 56.1V (for LFP: 3.3V × 13-17)
Max Pack Voltage / Operating Range	~58.4V (17S × 3.43V), typical: 45-58.4V
Input Power & Communication Power Requirements	9V-60V input, 12V typical, CAN-powered or separate DC
Cell Voltage & Balancing	
Cell Voltage Measurement Range	1.5V to 4.2V
Measurement Accuracy & Resolution	±5mV; 1mV resolution
Balancing Topology (active or passive/dissipative)	Passive
Balancing Current	~30-100 mA (depending on configuration)
Balance Enable/Threshold Voltages	Typically enabled >3.45V
Protections & Safety	
Over-voltage Protection (per cell and pack)	Cell: >3.65V; Pack: >58.8V
Under-voltage Protection (per cell and pack)	Cell: <2.5V; Pack: <36V
Over- & Under-temperature cut-offs	Charge: 0°C to 45°C; Discharge: -20°C to 60°C
Leakage/Ground fault detection	Optional (RCD/GFD integration)
Protection switching topology	MOSFET/Contactor based
Precharge circuit or inrush current limiting	Yes (resistive soft start/ inrush limiter)
Communication Interfaces	
Supported Protocols	CAN 2.0B (500kbps), UART (optional), BLE (ESP8266)
Baud Rates & CAN Speeds	CAN: 125k to 1 Mbps; UART: 9600–115200
General-purpose I/O	6 (Analog Inputs)
Host Interfaces	CAN or UART to host/ controller/charger
Standards & Certifications	
Safety Standards	AIS-038 Rev 2, IEC 62133-2
EMS/EMC compliance, regulatory notes	BIS applicable, CISPR 25 in roadmap

Current Handling	Parameters
Continuous Charge Current Limit	100A (based on 100Ah pack, 1C)
Peak Charge & Discharge Currents (duration and recovery conditions)	150A for <10s
Continuous Discharge and Peak Discharge Limits	100 A / 150 A (<10 s)
Internal Circuit Resistance	<5 mΩ (BMU + cabling)
Temperature Sensing & Thermal Management	
Number & Type of Temperature Sensors Supported	4-8 NTCs (10kΩ typical)
Sensor Range & Accuracy	±1°C
Thermal Protection Thresholds and cut-offs	-20°C to +60°C (pack operation); -0°C or >60°C triggers cutoff
Mechanical & Environmental	
Dimensions & Weight	Custom board; ~100mm × 70mm (est.)
Mounting Form Factor / Connector Types	Configurable
Operating & Storage Temperature Range	-20°C to +60°C ; -40°C to +70°C
Humidity / Environmental Classifications (indoor/ outdoor, shock & vibration specs)	10–90% RH, non-condensing
Ingress / IP rating (if enclosure provided)	IP20–IP65 depending on enclosure
Software, Diagnostics & Tools	
Configuration Software Tools	PC-based GUI or CLI over UART/CAN
Event / Error Logging & RTC capabilities	Timestamped fault log (non-volatile) ; Yes (battery-backed optional)
Diagnostics & Self-Test features	Self-test on boot, watchdog, CAN alive
Versioning & Firmware upgrade method	UART or CAN bootloader supported
Performance & Lifecycle	
Efficiency	>98%
Power Consumption (active / idle / sleep current)	200mW / 50mW / <5mW
Cycle Life Expectations (if gauging capacity)	2000+ cycles (depends on cell)

XBATTERY

Let's Power the Future, Together!

At XBattery, we're redefining the way energy is stored, managed, and delivered.
Whether you're building smarter EV systems, renewable energy grids, or industrial power solutions, our team is here to help you design reliable, efficient, and scalable battery systems that drive progress.
Let's connect and create the next generation of intelligent energy solutions.

Contact Us

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